LESSON 8.4

Write and Represent Inequalities

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write inequalities to represent real-world situations.

WHY THIS MATTERS Detective Navarro is narrowing down suspects. A witness says the suspect is at least 18 years old. The detective writes the inequality a ≥ 18, where a is the suspect's age. How does this help eliminate suspects?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Inequality *Spanish: Desigualdad*	A math sentence that compares two sides with <, >, ≤, or ≥.	x ≥ 18
Greater than *Spanish: Mayor que*	The > sign, showing one number is bigger.	8 > 3 — eight is larger than three
Less than *Spanish: Menor que*	The < sign, showing one number is smaller.	2 < 7 — two is smaller than seven
At least / At most *Spanish: Al menos / A lo más*	'At least' means ≥. 'At most' means ≤.	'At least 10 players' → p ≥ 10 (could be 10, 11, 12, ...); 'At most 5 tries' → t ≤ 5 (could be 5, 4, 3, ...)
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	the x in x ≥ 18
No more than *Spanish: No más de*	A phrase meaning less than or equal to (≤).	“No more than 8 people” means people ≤ 8.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A van can carry at most 8 riders. Write an inequality for r, the riders.

1. Key words: “at most.” Is 8 allowed? Yes.
2. “At most” means less than or equal to: r ≤ 8.
3. Test: 8 works, 5 works, 9 does not.

Answer: r ≤ 8

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Clue: 'A bag holds fewer than 10 items.' What does 'fewer than' mean?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The phrase _____ tells me the symbol is _____ .
- My inequality is _____ because _____ .
- The value _____ (is / is not) a solution because _____ .
- This inequality has many solutions because _____ .

Word bank inequality | at least | at most | more than | less than | greater than | symbol | constraint | variable | solution | boundary | range

WATCH OUT!

A common mistake

A common mistake in Write Inequalities is using a strict symbol (> or <) for a phrase that actually includes the boundary value. For 'the suspect is at least 18 years old,' students often write a > 18 instead of the correct a ≥ 18

LESSON 8.4

Write and Represent Inequalities

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Place markers on the number line for each value, then decide: is 5 > 3 true or false? Is 2 ≥ 7 true or false?



HINT On a number line, greater numbers are to the right. 5 is to the right of 3, so 5 > 3 is true.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The temperature is less than 40 degrees'?

(A) t < 40
(B) t > 40
(C) t ≤ 40
(D) t = 40

HINT Find the comparison phrase: 'less than 40 degrees'. Does 40 itself count, or only temperatures below it?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'You must be at least 48 inches tall to ride'?

(A) $h \geq 48$
(B) $h > 48$
(C) $h < 48$
(D) $h \leq 48$

HINT 'At least 48' — ask yourself: can a rider who is EXACTLY 48 inches tall ride? That decides whether 48 is included.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'A store can hold no more than 75 people'?

(A) $p \leq 75$
(B) $p < 75$
(C) $p \geq 75$
(D) $p > 75$

HINT 'No more than 75' — could the store hold exactly 75 people? Could it hold 76?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The speed limit is more than 25 mph'?

(A) $s > 25$
(B) $s \geq 25$
(C) $s < 25$
(D) $s = 25$

HINT 'More than 25' — is exactly 25 included, or only speeds above 25?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Detective's Mistake

1. **Clue** The witness says the suspect is at least 16 years old.
2. **Choose a variable** Let a stand for the suspect's age.
3. **Write the inequality** a > 16

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Does 'at least 16' allow the suspect's age to be exactly 16, or must it be older?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?

SENTENCE STARTER 'At least' uses ≥ instead of > because it means the value ___ be equal to the number, while 'more than' means it must be ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.4

Write and Represent Inequalities

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write inequalities to represent real-world situations.

WHY THIS MATTERS Detective Navarro is narrowing down suspects. A witness says the suspect is at least 18 years old. The detective writes the inequality $a \geq 18$, where a is the suspect's age. How does this help eliminate suspects?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each situation to its inequality

_____ Item 1 (A)

_____ Item 2 (B)

_____ Item 3 (C)

_____ Item 4 (D)

2 **Complete the table. Show how you found each value.**

Identify the key phrase, symbol, and write the inequality for each situation.

Situation	Key Phrase	Symbol	Inequality
A package weighs less than 50 pounds.			
You need at least $25 to buy the game.			
No more than 30 students can join the club.			

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Test x = 10 in each inequality. Sort by whether it works.

Groups: **x = 10 IS a solution** **x = 10 is NOT a solution**

_____ $x \geq 10$

_____ $x \leq 15$

_____ $x > 8$

_____ $x > 10$

_____ $x < 10$

_____ $x \leq 5$

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A parking garage charges if your stay is more than 2 hours. Which inequality represents times t that are charged?

(A) $t > 2$

(B) $t \geq 2$

(C) $t < 2$

(D) $t \leq 2$

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Crack the Inequality Error

1. **Clue** The report says no fewer than 9 witnesses saw the suspect.
2. **Choose a variable** Let w stand for the number of witnesses.
3. **Write the inequality** $w < 9$

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 8.4

Write and Represent Inequalities

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

For each situation, decide if it needs an equation or inequality and explain why.

Equation or Inequality?	Model	Why?
A detective has exactly 12 cases.		
A detective has more than 12 cases.		
A detective has at most 12 cases.		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Inequality Error

1. **Situation** No more than 25 students can join the field trip
2. **Identify key phrase** 'No more than' means greater than
3. **Write inequality** s > 25

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Write two different real-world situations: one that uses > and one that uses ≥. Explain why the situations need different symbols even though both mean 'bigger.'

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do the words 'at least' and 'at most' change the inequality symbol compared to 'more than' and 'fewer than'?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help